Fast cube roots in $\mathbb{F}_{p^2}$ via the algebraic torus

Youssef El Housni

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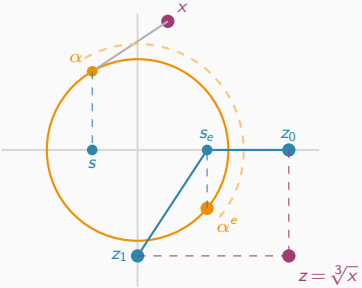
Consensys – Linea



Why $\mathbb{F}_{p^2}$ cube roots?

Applications of $\mathbb{F}_{p^2}$ cube roots

- **Hash-to-curve** (Icart and Koshelev maps [Ica09, Kos21b])

  Koshelev: "does not know how to express $\sqrt[3]{\cdot}$ in $\mathbb{F}_{p^2}$ through $\mathbb{F}_p$-roots"

- **Point decompression** on $j=0$ curves ($y^2 = x^3 + b$ over $\mathbb{F}_{p^2}$) / $\mathbb{G}_2$ subgroups
  - **Highly 2-adic fields (SNARK)**: cube root cheaper than square root
  - **Batch decompression (via quotient varieties [Kos21a])**: requires cube roots

- **3-isogeny walks**: bottleneck = one $\mathbb{F}_{p^2}$ cube root per step

3-isogeny walk

$$E_0 \xrightarrow{\sqrt[3]{\cdot}} E_1 \xrightarrow{\sqrt[3]{\cdot}} E_2 \xrightarrow{\sqrt[3]{\cdot}} \cdots \xrightarrow{\sqrt[3]{\cdot}} E_m$$

bottleneck: one $\mathbb{F}_{p^2}$ cube root per step

Application [NO24]	Security	Walk length	Field
QFESTA-128	128-bit	$m = 162$	$\mathbb{F}_{p^2}$
QFESTA-192	192-bit	$m = 243$	$\mathbb{F}_{p^2}$
QFESTA-256	256-bit	$m = 324$	$\mathbb{F}_{p^2}$

This talk

1. Reduce $\mathbb{F}_{p^2}$ cube root to $\mathbb{F}_p$ operations
   via the **algebraic torus** $\mathbb{T}_2(\mathbb{F}_p)$

2. **1.4–2.1**$\times$ faster than best prior (direct) method

Roots in $\mathbb{F}_p$

Square root in $\mathbb{F}_p$

- Simple cases via direct exponentiation:

  $p \equiv 3 \pmod 4$: $\sqrt{a} = a^{(p+1)/4}$, $\quad p \equiv 5 \pmod 8$: Atkin's formula, $\quad \ldots$

- General($v_2(p-1) \geq 3$): **Tonelli–Shanks**

  Exponentiation + discrete-log correction in 2-Sylow subgroup

- Cost $\approx 1.5 \log_2(p)\, \mathbf{M}_p$ (via square-and-multiply)
- Cost $\approx 1.2 \log_2(p)\, \mathbf{M}_p$ (via addition chain)

Cube root in $\mathbb{F}_p$

- Simple cases via direct exponentiation:

  $p \equiv 2 \pmod 3$: $\sqrt[3]{a} = a^{(2p-1)/3}$

  $p \equiv 4 \pmod 9$: $\sqrt[3]{a} = a^{(8p-5)/9}$

  $p \equiv 7 \pmod 9$: $\sqrt[3]{a} = a^{(p+2)/9}$

  $p \equiv 10 \pmod{27}$: $\sqrt[3]{a} = \zeta \cdot a^{(2p+7)/27}$

  $p \equiv 19 \pmod{27}$: $\sqrt[3]{a} = \zeta \cdot a^{(p+8)/27}$

- General ($v_3(p-1) \geq 3$): **Adleman–Manders–Miller [AMM77]**

  Exponentiation + discrete-log correction in 3-Sylow subgroup

- Cost $\approx 1.5 \log_2(p)\ \mathbf{M}_p$ (via square-and-multiply)
- Cost $\approx 1.2 \log_2(p)\ \mathbf{M}_p$ (via addition chain)

Square root vs. cube root in $\mathbb{F}_p$

v		**Square root** ($v_2(p-1)$)	**Cube root** ($v_3(p-1)$)
1	Formula	1 exp $(p \equiv 3 \bmod 4)$	1 exp $(p \equiv 4, 7 \bmod 9)$
2	Formula	1 exp $(p \equiv 5 \bmod 8)$	1 exp $(p \equiv 10, 19 \bmod 27)$
≥ 3	Algorithm	Tonelli–Shanks (DL in 2-Sylow)	AMM (DL in 3-Sylow)

All practical cryptographic primes have $v_3(p-1) \leq 2$

$\Rightarrow$ cube root in $\mathbb{F}_p$ = single exponentiation

From $\mathbb{F}_p$ to $\mathbb{F}_{p^2}$

$\mathbb{F}_{p^2}$ arithmetic recap

$$\mathbb{F}_{p^2} = \mathbb{F}_p[u]/(u^2 - \beta), \qquad \mathrm{N}(x) = x \cdot \bar{x} \in \mathbb{F}_p$$

- Conjugation: $\overline{x_0 + x_1 u} = x_0 - x_1 u$
- Frobenius: $x^p = \bar{x}$
- $\mathbf{M}_{p^2} = 3\,\mathbf{M}_p$ (Karatsuba)
- $\mathbf{S}_{p^2} = 2\,\mathbf{M}_p$ (Complex)

$\sqrt{\cdot}$ in $\mathbb{F}_{p^2}$ reduces to $\mathbb{F}_p$

Key identity: $b = x^{(p+1)/2}$ satisfies $b^2 = \mathrm{N}(x) \in \mathbb{F}_p$, so $b \in \mathbb{F}_p$

$$x \in \mathbb{F}_{p^2} \xrightarrow{x \cdot \bar{x}} \mathrm{N}(x) \in \mathbb{F}_p \xrightarrow{\mathbb{F}_p \text{ sqrt}} \sqrt{\mathrm{N}(x)} \in \mathbb{F}_p \xrightarrow{\text{lift}} z \in \mathbb{F}_{p^2}$$

$\Rightarrow$ $\mathbb{F}_{p^2}$ square root reduces to **one** $\mathbb{F}_p$ **square root** + cheap recovery

$\sqrt[3]{\cdot}$ in $\mathbb{F}_{p^2}$: why the same approach fails

Try $\gamma = x^{(p+1)/3}$: $\quad \gamma^3 = \mathrm{N}(x) \in \mathbb{F}_p \quad$ but $\gamma \notin \mathbb{F}_p$ in general!

$$x \in \mathbb{F}_{p^2} \xrightarrow{x \cdot \bar{x}} \mathrm{N}(x) \in \mathbb{F}_p \xrightarrow{\mathbb{F}_p \text{ cbrt}} \sqrt[3]{\mathrm{N}(x)} \in \mathbb{F}_p \overset{\text{✗ lift?}}{\dashrightarrow} z \in \mathbb{F}_{p^2}$$

Casus irreducibilis: the cubic for z_0 over $\mathbb{F}_p$ has no $\mathbb{F}_p$-radical solution

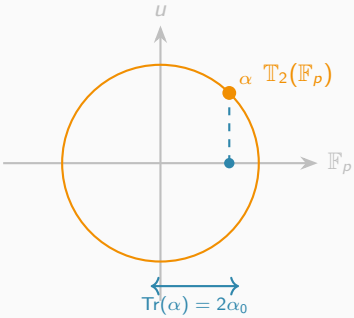
$\Rightarrow$ *Need a different approach*

Our algorithm

The algebraic torus $\mathbb{T}_2(\mathbb{F}_p)$

$$\mathbb{T}_2(\mathbb{F}_p) = \{\alpha \in \mathbb{F}_{p^2}^* : \mathsf{N}(\alpha) = 1\}, \qquad |\mathbb{T}_2| = p + 1$$

- $\alpha^p = \bar{\alpha} = \alpha^{-1}$
- α determined by $\mathsf{Tr}(\alpha) = 2\alpha_0 \in \mathbb{F}_p$



Key observation

1. $x^{p-1} \in \mathbb{T}_2(\mathbb{F}_p)$ (since $\mathrm{N}(x^{p-1}) = 1$)
2. $\gcd(3, p+1) = 1$ for $p \equiv 1 \pmod 3$

   $\Rightarrow$ **cube map is a bijection** on $\mathbb{T}_2(\mathbb{F}_p)$
3. Cube root on torus $=$ exponentiation by $e = 3^{-1} \bmod (p+1)$
4. **Connection to Lucas V-sequence:**

   $\alpha \in \mathbb{T}_2$ is determined by $s = \mathrm{Tr}(\alpha) \in \mathbb{F}_p$, and

   $$\mathrm{Tr}(\alpha^e) = V_e(s) \quad \text{(Lucas identity)}$$

   $\Rightarrow$ compute α^e via $V_e(s)$ **entirely in** $\mathbb{F}_p$

$$V_e\big(\mathrm{Tr}(\alpha)\big) = \mathrm{Tr}(\alpha^e), \qquad e = 3^{-1} \bmod (p+1)$$

Algorithm overview

projection to $\mathbb{T}_2(\mathbb{F}_p)$

exponentiation in $\mathbb{T}_2(\mathbb{F}_p)$

recovery

$$x = x_0 + x_1 u \in \mathbb{F}_{p^2}$$

$$\alpha = \mathrm{Tr}(x^{p-1}) = 2(x_0^2 - |\beta| x_1^2) \cdot N(x)^{-1}$$

$$s = V_e(\alpha)$$
$$e = 3^{-1} \bmod (p+1)$$

$$z_0 = \frac{x_0}{\sqrt[3]{\mathrm{N}(x)}\,(s-1)}$$
$$z_1 = \frac{x_1}{\sqrt[3]{\mathrm{N}(x)}\,(s+1)}$$

$$z = z_0 + z_1 u \in \mathbb{F}_{p^2}$$

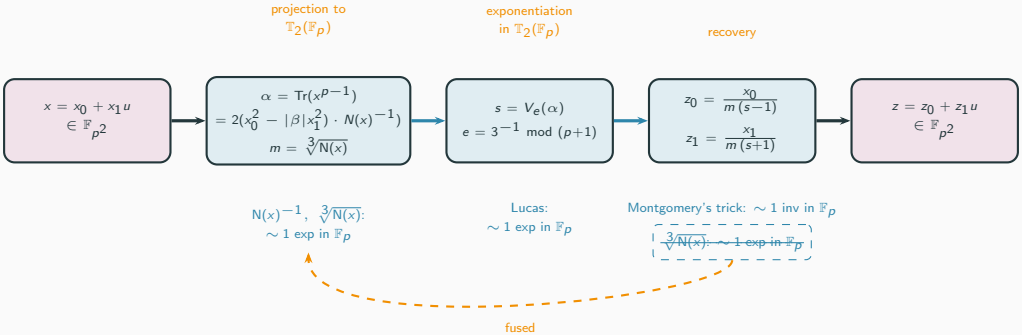
$\mathrm{N}(x)^{-1}$: ~ 1 inv in $\mathbb{F}_p$

Lucas: ~ 1 exp in $\mathbb{F}_p$

$\sqrt[3]{\mathrm{N}(x)}$: ~ 1 exp in $\mathbb{F}_p$

$x_0 / \cdot$, $x_1 / \cdot$: ~ 2 inv in $\mathbb{F}_p$

Algorithm overview (optimized)



Fused $\sqrt[3]{N(x)}$ and $N(x)^{-1}$

Let $n = \mathrm{N}(x)$. Write $p - r = d \cdot 3^{v_3(p-1)}$ with $\gcd(d, 3) = 1$ and $r \in \{1, 4, 7\}$.

A single addition chain computes $t = n^{(p-r)/d}$, then:

$$m = n \cdot t^s \quad \text{(cube root)}, \qquad n^{-1} = m^{r-2} \cdot t^{d-(r-2)s} \quad \text{(Fermat)}$$

where $s = (1 - r)/3 \bmod d$.

Prime	p mod 9	$(r,\ s)$	Saves
p-p377	7	(7, 4)	$\sim 80\,\mathbf{M}_p$ ($\approx$ 1 inversion)
i-p381	4	(4, 5)	

$\Rightarrow$ Cube root **and** inverse of $\mathrm{N}(x)$ from one addition chain

Lucas V-sequence on the torus

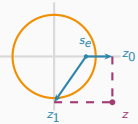
$$V_0 = 2, \quad V_1 = s$$

$$V_{2k} = V_k^2 - 2, \qquad V_{2k+1} = V_k \cdot V_{k+1} - s$$

- Cost: $2\,\mathbf{M}_p$ per bit (1 mul + 1 sqr)
- Binary ladder: tight, branch-free, constant-time
- Total: $\approx 2\ell\,\mathbf{M}_p$ where $\ell = \log_2(p)$

Lucas dominates cost: 55–60% of total

Recovery & verification



Recovery:

$$z_0 = \frac{x_0}{m(s-1)}, \qquad z_1 = \frac{x_1}{m(s+1)}$$

Single $\mathbb{F}_p$ inversion via Montgomery's trick: $5\,\mathbf{M}_p + 1\,\mathbf{I}_p$

Verify and adjust:

- Check $z^3 = x$; if not, multiply by ζ_3 or ζ_3^2
- $\zeta_3 \in \mathbb{F}_p \Rightarrow$ multiplication costs only $2\,\mathbf{M}_p$
- Cost $\leq 6\,\mathbf{M}_p$

Cost comparison ($\ell = \log_2 p$)

Method	Cost	Ratio
Direct (addition chain in $\mathbb{F}_{p^2}$)	$\sim 5.3\ell\ \mathbf{M}_p$	$1.7\times$
Frobenius 1-bit	$\sim 4.25\ell\ \mathbf{M}_p$	$1.3\times$
Frobenius 2-bit	$\sim 3.4\ell\ \mathbf{M}_p$	$1.06\times$
Torus	$\sim 3.2\ell + 100\ \mathbf{M}_p$	$\mathbf{1}\times$

Structural advantage: all operations in $\mathbb{F}_p$ ($3\times$ cheaper than $\mathbb{F}_{p^2}$)

Benchmarks

Target primes

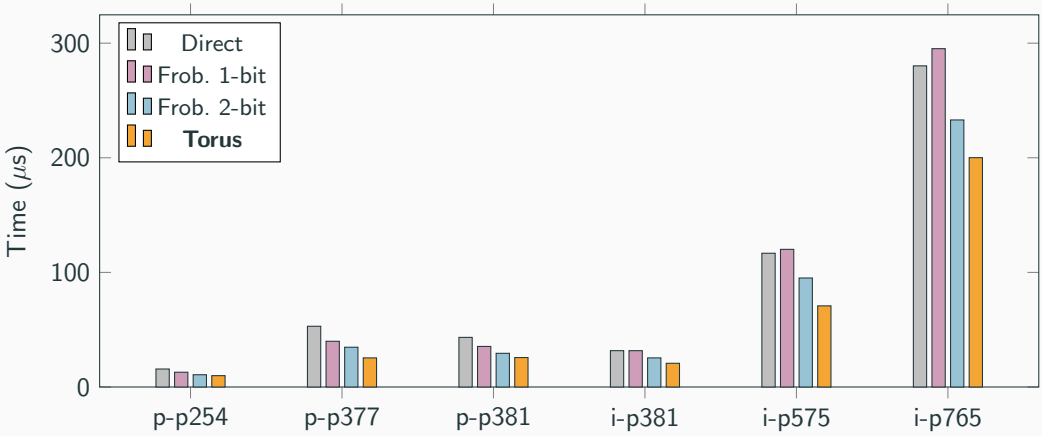
Prime	$\log_2 p$	$p \bmod 9$	β	**Source**
p-p254	254	1	-1	Pairing
p-p377	377	7	-5	Pairing
p-p381	381	1	-1	Pairing
i-p381	381	4	-1	Isogeny
i-p575	575	4	-1	Isogeny
i-p765	765	4	-1	Isogeny

Benchmark results

Apple M5, Go 1.24, median of 4 runs

Prime	Direct (μs)	Frob. 1 (μs)	Frob. 2 (μs)	Torus (μs)	vs. Direct	vs. Frob. 2
p-p254	15.7	12.9	10.7	**9.9**	1.59×	1.08×
p-p377	53.0	39.9	34.7	**25.4**	2.09×	1.37×
p-p381	43.3	35.4	29.4	**25.7**	1.69×	1.14×
i-p381	31.7	31.7	25.4	**20.7**	1.53×	1.23×
i-p575	116.7	120.1	95.1	**70.8**	1.65×	1.34×
i-p765	280.2	295.2	233.0	**200.1**	1.40×	1.16×

Visual comparison



Constant-time & conclusion

Constant-time considerations

All methods: fixed operation sequence, CT w.r.t. input

VerifyAndAdjust: input-dependent branch (multiply by ζ_3^k)

Application	**Secret input?**	**CT needed?**
Hash-to-curve	Yes	Yes
Point decompression	No	No
3-isogeny walk (CSIDH)	Yes	Yes

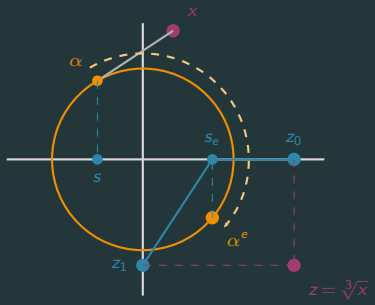
CT VerifyAndAdjust via conditional select ($\sim 6\,\mathbf{M}_p$ overhead)

Conclusion

- $\mathbb{F}_{p^2}$ cube root $\longrightarrow \mathbb{F}_p$ via **algebraic torus** + **Lucas sequences**
- **1.4–2.1**$\times$ vs. direct, **1.1–1.4**$\times$ vs. Frobenius 2-bit
- Extension to $p \equiv 2 \pmod 3$ discussed in paper
- Extension to $\mathbb{F}_{p^{2k}}$ + GLV-like optimization discussed in paper
- Code: `github.com/yelhousni/fp2-cbrt`

Thank you!

github.com/yelhousni/fp2-cbrt



References i

Leonard M. Adleman, Kenneth Manders, and Gary L. Miller.
On taking roots in finite fields.
18th Annual Symposium on Foundations of Computer Science (FOCS), pages 175–178, 1977.

Thomas Icart.
How to hash into elliptic curves.
In Shai Halevi, editor, *CRYPTO 2009*, volume 5677 of *LNCS*, pages 303–316. Springer, Berlin, Heidelberg, August 2009.

Dmitrii Koshelev.
Batch point compression in the context of advanced pairing-based protocols.
Cryptology ePrint Archive, Report 2021/1446, 2021.

References ii

Dmitrii Koshelev.
Some remarks on how to hash faster onto elliptic curves.
Cryptology ePrint Archive, Report 2021/1082, 2021.

Kohei Nakagawa and Hiroshi Onuki.
QFESTA: Efficient algorithms and parameters for FESTA using quaternion algebras.
In Leonid Reyzin and Douglas Stebila, editors, *CRYPTO 2024, Part V*, volume 14924 of *LNCS*, pages 75–106. Springer, Cham, August 2024.